Identification of Novel Cryptic Pockets in RAC1

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Abstract

The work described here attempts to locate cryptic pockets in the RAC1 protein. More to be written

Introduction

- importance of GTPases, their switch mechanism and their interactions with GEFs, GAPs and GDIs Small GTPAses act as molecularswitches cycling between OFF and ON states. These switching mechnism thus controls a vast variety of processes that control cell growth and differentiation to vesicular and nuclear transport.
- structural biology information about RAC1 and protein-protein interactions with GEFs, GAPs and GDIs
- Therapeutic interventions on RAC1 and their disadvantages
- Binding site information and changes in the binding sites, if any are reported
- Are there any reports on cryptic pockets for RAC1

Methods

- List of PDBS used
- Protocol for AMBER MD simulations
- Protocol for POVME2
  - pocket identification
  - pocket volume measurements
- fragment placement and trying yo keep the pocket open to target drug-like molecules
  - Type of pocket and the type of fragment more suitable to keep it open
- Analysis of other MD observations to support pocket volume changes
  - RMSDs for switch region 1 and 2
  - (perhaps) PCA analysis to explain the switch region dynamics and the pocket volumes

Results and discussion

- Try to look for a connection/relation between the following
  - switch region dynamics with different binding partners
  - switch region dynamics with pocket volumes
  - If we find different pockets that might open and close with different binders

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